Gogula Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.76
APSWR SCHOOL/Jr COLLEGE FOR BOYS,GANDIKSHETRAM	Class XII (Board of Intermediate Education, Andhra Pradesh (BIEAP))	2020	9.88%
ZPHS TEKURPET, Y.S.R DISTRICT	Class X (Board of Secondary Education, Andhra Pradesh)	2018	9.30%

WORK EXPERIENCE

ITProfound | Full Stack Developer Intern

- Built full-stack web applications using **React.js, Node.js, Express.js, and MongoDB**, developing responsive interfaces and backend services for complete application workflows.
- Engineered **RESTful APIs** with Node.js and Express.js, enabling reliable frontend-backend communication.
- Designed reusable **React.js components** with modular architecture, improving **maintainability, reusability, and UI consistency** across application interfaces.
- **Optimized MongoDB data operations and backend workflows** by structuring efficient data-access patterns, improving **data retrieval efficiency and application reliability**.

PROJECTS

Distributed Flight Booking & Reservation System GitHub

- Architected a flight booking platform into **6 microservices** behind an API Gateway using **Docker, Nginx, RabbitMQ, and database-per-service**, enabling independent deployment and asynchronous workflows.
- Engineered concurrency-safe booking with **PostgreSQL row-level locking and idempotency keys**, preventing **seat overselling and duplicate requests** across booking, payment, cancellation, and refund workflows.
- Optimized flight search with **Redis caching** and **3 stateless API Gateway replicas** behind Nginx, achieving **945 RPS at 102 ms P95 latency** under **50 concurrent connections** with **0% reported errors**.

AI-Powered Semantic E-Commerce & Recommendation Platform GitHub

- Developed an **AI-powered e-commerce platform** using Next.js, MongoDB, Firebase, Stripe, and JWT, supporting secure authentication, product discovery, cart management, and end-to-end order processing.
- Built **semantic product search and recommendations** using Sentence Transformers (all-MiniLM-L6-v2) and FAISS for embedding-based Top-K retrieval and meaning-based product discovery.
- Integrated a **DistilBERT sentiment analysis pipeline** for customer reviews and connected AI services through REST APIs, with role-based access control, Stripe payments, and MongoDB-backed cart and order workflows.

Intelligent Job Recommendation & Matching System GitHub

- Architected a job recommendation platform using **React, Node.js/Express, PostgreSQL, and a Python Flask ML microservice** with **Docker Compose**, enabling independent backend and ML service development.
- Improved personalized job matching using **TF-IDF and cosine similarity**, achieving **60% Precision@5 and 100% Recall@5** across **3 users and 10 jobs**, with Top-K ranking and matched-skill explanations.
- Secured backend workflows with **JWT, bcrypt, parameterized PostgreSQL queries, pagination, and application constraints**, with TF-IDF delivering a **12.51% Precision@5 improvement** over keyword matching.

SKILLS

Languages: C/C++, Java, JavaScript, Python, SQL

Frontend: HTML, CSS, React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, Microservices, JWT, RBAC, bcrypt

System Design: HLD, LLD, OOP, SOLID Principles, Design Patterns, System Architecture, API Design

Databases: PostgreSQL, MySQL, MongoDB, Redis

AI/ML: Machine Learning, Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Deep Learning, Neural Networks, CNNs, RNNs, NLP

DevOps & Testing: Docker, Docker Compose, Git, GitHub, Jest, Postman

RELEVANT COURSES

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Computer Organization & Architecture, Machine Learning, Deep Learning

POSITIONS OF RESPONSIBILITY

Coding Club Coordinator

- **Coordinated** coding activities for **150+ junior students**, supporting programming and problem-solving skills.

Web Development Competition Coordinator | College Fest

- **Led** a web development competition for **100+ students** from multiple colleges across districts.

SCHOLASTIC ACHIEVEMENTS

- **Competitive Programming:** Solved **1000+ DSA problems** across **LeetCode and GeeksforGeeks**.
- **HackerRank:** Earned **5-star badges** in programming and problem solving.
- **NMMS Scholarship:** Qualified for the **National Means-cum-Merit Scholarship (NMMS)** examination.

EXTRACURRICULAR

Fitness: Gym | **Sports:** Cricket, Volleyball | **Cultural:** Dancing